PHYS-467 Exercise: Revisiting Inference

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Exercise 1: Particle Decay

Recall the particle decay problem you saw in the lecture. We have a source that emits particles, which decay at some point. The probability of a particle decay at position x is distributed according to

$$P(x) = \frac{1}{\lambda} \exp\Big(-\frac{x}{\lambda} \Big) . \tag{1}$$

Our goal is to infer the λ^{true}, the "ground truth" value of λ, given some data points x that were obtained from the above distribution. The dataset contains the exact x_i (suppose we have an infinite detector), i.e. we use data points $\underline{x} = \{x_i\}_{i=1,\ldots,N}$ with

$$x_i \sim P(x_i|\lambda^{\text{true}}) = \frac{1}{\lambda^{\text{true}}} e^{-x_i/\lambda^{\text{true}}} \quad \text{i.i.d.} \tag{2}$$

In this exercise, we want to construct the maximum likelihood and MAP estimators for finding λ^{true}.

1. Write the likelihood for a complete dataset $\underline{x}$, i.e. $\mathcal{L}(\lambda) = P(\underline{x}|\lambda)$.
2. Write the posterior distribution $P(\lambda|\underline{x})$ in terms of the prior and the likelihood.
3. Compute the maximum likelihood estimator $\hat{\lambda}_{\text{ML}} = \arg\max_\lambda \log P(\underline{x}|\lambda)$.
4. Let's assume we have some prior information about the distribution of the λ, e.g. the exponential prior $P(\lambda) = -\frac{1}{\lambda_0} e^{-\frac{\lambda}{\lambda_0}}$, where λ_0 is the mean value of this distribution. Compute the maximum aposterior estimator $\hat{\lambda}_{\text{MAP}} = \arg\max_\lambda \log[P(\underline{x}|\lambda)P(\lambda)]$ in terms of the data and the prior λ_0. What happens to the expression when we increase the number of data samples as $N \to \infty$?

Exercise 2: Bent Coin

We now want to look at (un)fair coin tosses. Imagine that we have a coin and the probability that the coin shows the head is p, and $1-p$ for tail, a Bernoulli distribution. Let $\underline{s} = (s_1, s_2, \ldots, s_N)$ be a sequence of tosses that you observed from the same coin and let $s_i = 1$ mean that a head was tossed in the i^{th} trial.

1. We now want to determine if the coin was fair, i.e. whether $p = 1/2$ or not, for which we need to find an estimator for p. For this, compute the likelihood $P(\underline{s}|p)$ and derive the maximum likelihood estimator $\hat{p}_{\text{ML}}$.
2. You take part in a game where you observe the first 7 tosses of a possibly bent coin, as $\underline{s} = (1, 1, 0, 0, 1, 1, 1)$. As part of the game, you have to decide whether you want to bet on the next two tosses being head. If you are correct you win the game. To make an informed decision on your bet, evaluate $P(s_8 = 1, s_9 = 1|\hat{p}_{\text{ML}})$ and determine what you should bet on.

3. Assume that you obtained knowledge on the game that the coins are usually not bent uniformly in the range of $[0, 1]$, but are far more likely to be bent close to a fair coin. You model this prior as $P(p) = 6p(1 - p)$ and additionally add that $p \in [0, 1]$. Take a Bayesian approach to your bet and compute

$$\int P(s_8 = 1, s_9 = 1|p)P(p|\underline{s})dp$$

to determine what you want to bet on.